A transitive input to the rank and degree extension

Human Compression project

Private bounded correction; distinct review pending

Abstract

We construct a rank-five string C-group acting as S_{11}, with a perfect split satisfying the hypotheses of the rank and degree extension in the specified Cameron–Fernandes–Leemans author edition. Its prescribed extension fails the intersection property. The input C-property is proved through explicit subgroup formulas and the standard end criterion. Thus no finite census is a premise of the counterexample. The example is below the rank range of the source's main stability theorem.

1 The marking and the source claim

On $\{1, \dots, 11\}$ define

$$x = (8\ 9)(10\ 11), \quad y = (1\ 8)(9\ 10),$$
$$q = (1\ 2)(3\ 4), \qquad r = (2\ 6)(3\ 5)(4\ 7), \quad s = (3\ 4)(6\ 7).$$

The ordered marking is (x, y, q, r, s). All five permutations are distinct nonidentity involutions; distant generators commute. Composition below uses right actions, so $z^{ab} = (z^a)^b$ and $a^b = b^{-1}ab$. For a supported permutation g, write $\varepsilon(g) \in \mathbb{F}_2$ for its parity.

Our exact source is Cameron–Fernandes–Leemans, *The number of string C-groups of high rank*, arXiv:2212.12723v2, 11 January 2023 (L019). Its Theorem 6.4, pp. 32–33, concerns a C-marking with a perfect split, a fracture graph, both split endpoints moved by the deleted subgroup, and one trivial internal factor of the split generator. It asserts that replacing that generator by the prescribed two generators at a new point preserves the C-property. We verify the printed definitions on pp. 8–9 and 16–17 directly after proving that this input is C. We retain the standard end-parabolic criterion, source Proposition 2.1, attributed there to McMullen–Schulte, Proposition 2E16. Every subgroup formula used with that criterion is proved below.

2 A small permutation calculation and a parity character

Put $t = (3\ 4)$, $u = (3\ 5)(4\ 7)$, $D = \langle t, u \rangle$ on $Y = \{3, 4, 5, 7\}$, and $\tau = (9\ 10)$. The product tu is a 4-cycle, so D is dihedral of order eight. There is a homomorphism $\chi : D \longrightarrow \mathbb{F}_2$ with $\chi(t) = \chi(u) = 1$: its kernel is the cyclic rotation subgroup $\langle tu \rangle$. This is a character of the dihedral marking, not necessarily the ordinary parity of its action on Y.

Lemma 1. *The marking (q, r, s) is a rank-three C-marking of* $\mathrm{Sym}(\{1, 2, 3, 4, 5, 6, 7\})$.

Proof. Direct multiplication gives the short word

$$w = qr(qsr)^4 s = (4\ 5).$$

For clarity, the following complete table lists the conjugates needed:

z	1	q	r	rs	rsr	$rsrq$
w^z	(4 5)	(3 5)	(3 7)	(4 6)	(2 7)	(1 7).

Each equality is checked on the seven displayed points. The six edges in the bottom row form a spanning tree. Transpositions on the edges of a connected graph generate the whole symmetric group: along a path one obtains its endpoint transposition by successive conjugation. Hence $\langle q, r, s\rangle = S_7$.

On $\{1, 2, 6\}$ the generators q, r act as (1 2), (2 6); on Y they act as t, u. Consequently

$$T := \langle q, r\rangle = \{gd : g \in \mathrm{Sym}(\{1, 2, 6\}),\ d \in D,\ \varepsilon(g) = \chi(d)\}. \tag{1}$$

Indeed the group is contained in this set, of order 24, while qr has order $\mathrm{lcm}(3, 4) = 12$. Two involutions whose product has order 12 generate a dihedral group of order 24, proving equality. The product rs is the cycle (2 7 3 5 4 6), so $R := \langle r, s\rangle$ is the dihedral group of order 12 in its natural six-point action. An element of $T \cap R$ fixes 1 and preserves $\{1, 2, 6\}$, hence preserves the adjacent pair $\{2, 6\}$ of this six-cycle. Its setwise stabilizer in the dihedral group is $\langle r\rangle$, of order two: no nonidentity rotation preserves an edge, and precisely one reflection interchanges its endpoints. Thus $T \cap R = \langle r\rangle$. The two rank-two end markings are C, and the standard end criterion proves the lemma. □

We will also use the elementary following observation. If a transitive permutation group L on a set V fixes a new point b, then adjoining $(a\ b)$ for $a \in V$ generates $\mathrm{Sym}(V \cup \{b\})$. Conjugation by L gives every $(v\ b)$, and these generate that symmetric group. This observation is applied to restriction images below; it does not assume that the corresponding supported transpositions have already been separated in the original group.

3 The first rank-four end

Let

$$A = \langle x, y, q, r\rangle, \quad N = \langle x, y, q\rangle, \quad M = \langle y, q, r\rangle, \quad P = \langle y, q\rangle.$$

The two generators x, y form the natural dihedral group of order ten on $V = \{1, 8, 9, 10, 11\}$: their product is the 5-cycle (1 8 10 11 9). They are even permutations. The projection of N onto $V \cup \{2\}$ is therefore S_6, by adjoining the transposition (1 2) as in the preceding observation. Since the only action on Y is the factor t of q, one obtains

$$N = \{gt^{\varepsilon(g)} : g \in \mathrm{Sym}(\{1, 2, 8, 9, 10, 11\})\}. \tag{2}$$

The parity condition is respected by each generator; surjectivity of the projection and uniqueness of its permitted second coordinate prove the formula. Similarly,

$$P = \{gt^e\tau^v : g \in \mathrm{Sym}(\{1, 2, 8\}),\ \varepsilon(g) = e + v\}. \tag{3}$$

The projection is S_3 and $(yq)^3 = t\tau$, so it contains both points of each permitted fiber. Its order is 12. The group P fixes 11, while the stabilizer of 11 in the natural $D_{10} = \langle x, y\rangle$ is $\langle y\rangle$. Thus $\langle x, y\rangle \cap P = \langle y\rangle$, and the end criterion makes N a C-marking.

Next put $Z = \{1, 2, 6, 8\}$. Then

$$M = \{gd\tau^v : g \in \mathrm{Sym}(Z),\ d \in D,\ \varepsilon(g) = \chi(d) + v\}. \tag{4}$$

To prove surjectivity onto $\mathrm{Sym}(Z) \times D$, omit the τ coordinate. The images of y, q, r on Z are the three adjacent transpositions on the path $8, 1, 2, 6$, so their projection is S_4. Moreover y projects to a pure transposition in this product, with trivial D coordinate. Conjugating it by elements projecting to all of S_4 yields every transposition in $S_4 \times 1$. Cancelling these first coordinates then recovers both t, u in the second factor. The image is the full product. The displayed parity equation holds for all three original generators and uniquely determines v, proving (4).

The end subgroups of this marking are P and T from (1). A common element has its root permutation supported on $\{1, 2\}$. Membership in T forces $v = 0$, while membership in P forces its D coordinate to be t^e with $e = \varepsilon(g)$. Hence $P \cap T = \langle q\rangle$, proving that M is C.

The full first end has formula

$$A = \{gd : g \in \mathrm{Sym}(X),\ d \in D,\ \varepsilon(g) = \chi(d)\}, \qquad X = \{1, 2, 6, 8, 9, 10, 11\}. \tag{5}$$

Its projection onto X is S_7: the preceding S_6 projection becomes S_7 on adjoining the image (2 6) of r. The parity equation holds for all generators. In addition, $(qr)^4$ is a pure 3-cycle on $\{1, 2, 6\}$, since its D coordinate is $(tu)^4 = 1$. Its conjugates under the surjective S_7 projection are every 3-cycle on X, with trivial D coordinate. They generate $\mathrm{Alt}(X)$. Since the projection onto D is onto, multiplying any lift of d by these even permutations gives every root permutation of the required parity. This proves (5), without an assumption about unknown normal subgroups.

Finally (2) and (4) show $N \cap M = P$. Indeed a common permutation fixes 6 and 11, has an arbitrary root permutation g on $\{1, 2, 8\}$, a possible factor τ^v on $\{9, 10\}$, and a D coordinate restricted to t^e. Both formulas then give the same equation $\varepsilon(g) = e + v$, which is exactly (3). Thus both end markings of A are C and meet in the required middle group, so A is C.

4 The second end and the whole input

Put $B = \langle y, q, r, s\rangle$ and $Q = \langle q, r, s\rangle = S_7$. The projection of B onto $\{1, \dots, 8\}$ is S_8. Since $(1\ 2) \in Q$ and

$$(y(1\ 2))^3 = \tau,$$

we obtain

$$B = \mathrm{Sym}(\{1, \dots, 8\}) \times \langle\tau\rangle. \tag{6}$$

The end markings of B are M and Q, both already proved C. Their intersection fixes $8, 9, 10, 11$, so (4) reduces to $g \in \mathrm{Sym}(\{1, 2, 6\})$, $v = 0$ and $\varepsilon(g) = \chi(d)$. This is precisely $T = \langle q, r\rangle$. Therefore B is C.

For $A \cap B$, formula (6) forces a permutation in the X coordinate of (5) to preserve $Z = \{1, 2, 6, 8\}$, preserve $\{9, 10\}$, and fix 11. Write it as $g\tau^v$ with $g \in \mathrm{Sym}(Z)$. Its parity equation in (5) becomes $\varepsilon(g) + v = \chi(d)$, exactly the equation defining M. Thus $A \cap B = M$. The end criterion proves that (x, y, q, r, s) is a string C-marking.

Its full group is S_{11}, not merely a transitive subgroup. Formula (6) supplies the supported S_8 and $\tau = (9\ 10)$. Conjugating $(1\ 8) \in S_8$ by x gives $(1\ 9)$, which yields S_9. Adjoining τ yields S_{10}. Cancelling $(8\ 9)$ from $x = (8\ 9)(10\ 11)$ now yields $(10\ 11)$ and hence S_{11}.

5 The split, the fracture graph and the failed extension

For deletion of labels $0, 1, 2, 3, 4$, respectively, the orbit partitions are

0	$\{1, 2, 3, 4, 5, 6, 7, 8\}, \{9, 10\}, \{11\}$
1	$\{1, 2, 3, 4, 5, 6, 7\}, \{8, 9\}, \{10, 11\}$
2	$\{1, 8, 9, 10, 11\}, \{2, 3, 4, 5, 6, 7\}$
3	$\{1, 2, 8, 9, 10, 11\}, \{3, 4\}, \{5\}, \{6, 7\}$
4	$\{1, 2, 6, 8, 9, 10, 11\}, \{3, 4, 5, 7\}$.

These partitions follow either from the formulas above or directly from connectivity in the deleted permutation graphs. Each has more than the one orbit of the full group. The respective edges $\{8, 9\}, \{1, 8\}, \{1, 2\}, \{2, 6\}, \{6, 7\}$ join distinct orbits in their rows, and form a path with isolated vertices added. Thus they are a fracture graph in the printed definition.

Take the perfect split at label $i = 2$, with $a = 1$, $b = 2$, $A_0 = \{1, 8, 9, 10, 11\}$ and $B_0 = \{2, 3, 4, 5, 6, 7\}$. Deleting q gives exactly these two orbits. The unique crossing edge is $\{1, 2\}$; the other edge $\{3, 4\}$ lies inside B_0. Both earlier marks are supported on A_0, both later marks on B_0, and the

split factorization is $q = (1\ 2)\beta$ with $\alpha = 1$ and $\beta = (3\ 4)$. The two endpoints move under y and r, respectively. All hypotheses of source Theorem 6.4 are now verified, including transitivity.

At the new point $c = 12$ the prescribed marking is

$$(x, y, u_0, v_0, r, s), \qquad u_0 = (1\ 12), \quad v_0 = (2\ 12)(3\ 4).$$

It satisfies the string commutations. However $(u_0 v_0)^3 = (3\ 4) = \beta$. The group $\langle v_0, r, s\rangle$ is the copy of the S_7 in Lemma 1 obtained by renaming point 1 as 12. Consequently it contains β. Equivalently the same word w, with q replaced by v_0, gives (4 5); conjugation by v_0 gives (3 5), and these give (3 4). But $\beta \notin \langle v_0\rangle$, since it is nonidentity and fixes 12, whereas v_0 moves 12. Thus the subgroups on labels $\{2, 3\}$ and $\{3, 4, 5\}$ meet in more than the subgroup on their common label $\{3\}$. The new marking is not C.

6 Limits of the conclusion

This refutes the printed Theorem 6.4 even for transitive inputs. It does not refute the main stability theorem: this example has rank $r = 5$ and degree $n = 11$, so it does not meet the stability range $r \geq (n+3)/2 = 7$. No conclusion about that remaining high-rank case, a corrected sufficient condition, full L019 recovery or historical priority is asserted here. The separate frozen disconnected example and the valid forward endpoint extension used in L010 remain unchanged. The computations supplied with this note are diagnostics of its explicit permutation and subgroup proof, not a substitute for the written argument.